AAE 535
Propulsion Design, Build, Test
Spring 2026

Objectives

The goals of this course are as-follows:

(i) To deepen student knowledge of topics relevant to general design processes as well as propulsion-specific considerations

(ii) To identify and formulate the design requirements for all components of a propulsion system

(iii) To describe a design that is likely to meet the set of requirements

(iv) To use modern engineering tools to complete the design process

(v) To apply engineering knowledge and skills acquired in previous classes to the design problem

(vi) To solve problems as part of a team

(vii) To effectively communicate by giving oral presentations and writing technical reports required of design engineers

Prerequisites

This course requires strong undergraduate and graduate preparation in the thermal sciences; including thermodynamics, compressible aerodynamics, and heat transfer. Prior exposure to material on engines or propulsion systems is preferable, but not required; co-enrollment in AAE 538 or AAE 539 is acceptable.

Recommended References

Humble, Henry, and Larson, *Space Propulsion Analysis and Design*, 1995, McGraw Hill.
Sutton and Biblarz, *Rocket Propulsion Elements*, 2001, John Wiley & Sons.
Use Compendex or Web of Science (https://www.lib.purdue.edu/find/databases) for literature related to your specific project.
* *Many, more specific references will be provided throughout the semester on the class shared drive.*

No textbook is required for this course.

Course Information and Access

Brightspace will be used as the primary mode through which course-related administrative content is shared from the instructor to the students (all sections, one course page). Grades will also be recorded and shared through this site.

A network drive has also been configured for this course. You can map the drive from any machine on the Purdue network: \\roger.ecn.purdue.edu\aae535. Authenticate as ecn*username* and use your career account password. Students on each team will have read/write/delete access within their project folder to enable collaboration and coordination. The instructor will share pertinent files in a specified directory. Each team must maintain an organized structure for their files/folders as this will be a part of the grade.

Students in this course will also be granted access to an ANSYS license with full meshing and physical simulation packages. You may use AAE computing servers to run large simulations that make use of this capability.

Assignments and Evaluations

The major deliverables in this class are as follows:

1. A Preliminary Design Review (PDR) presentation and written report that outlining the design requirements, a solution concept, and a budget that demonstrates understanding of existing designs, presents suitable justification that the proposed design meets system requirements with acceptable risk, and a contains a well-considered verification plan.
2. A Critical Design Review (CDR) presentation and accompanying written report that includes full documentation of the design, as well as a description of how the systems were analyzed and how they will be manufactured.
3. A Test Readiness Review (TRR) presentation with accompanying documentation to support testing operations.
4. A final report that includes test data and a comparison to design predictions. Report should be formatted as an AIAA conference paper.
5. A project comprehensive final video outlining the design concept, execution of the build process throughout the semester, testing and final outcomes.

Team grades will be assigned based on evaluations relative to pre-defined rubrics for the presentations and reports. Individual grades for major deliverables will be adjusted, relative to the team grade, based on feedback from peer evaluations submitted with each major deliverable. As the semester proceeds, time available may dictate some changes to the deliverables.

In addition to the major deliverables, a progress score contributes a portion towards the final course grade. These graded progress items will be in the form of short update meetings and state of workspaces, both physical and virtual. Specifically, the instructor will assign a progress score based upon the clarity of the presentations, organization of the files, and the state of the laboratory. All students will be expected to present multiple times. The progress score will include a regular assessment by the instructor. This assessment will include:

1. Verification after each design review that all individual design drawings and all design assemblies are saved in their native formats.
2. All presentations and Word documents are saved and clearly marked as final versions.
3. All purchase orders and associated quotes are saved and organized logically.
4. Photos and videos taken regularly throughout the semester and saved and organized logically.
5. All test documents (test data sheets, assembly and test procedures, test data, etc.) are saved and organized logically.
6. The laboratory build and test spaces are clearly organized and all parts are stored in designated locations.

Each of these deliverables and progress items will be graded by both the instructor and teaching assistant. All major design deliverable presentations and reports will be evaluated according to criteria listed on dedicated grading sheets (to be provided). Unless otherwise specified, all assignments will be due at the beginning of class. No late assignments will be accepted under any circumstance. Teams that do not complete a sufficient level of testing by the end of the semester will receive an incomplete for the class. The grading sheets will be completed by the instructor, any invited visitor the day of the presentation, and by all students not presenting. The instructor will still retain ultimate authority over assigning the final grades.

PDR Presentation and Document:	20%
CDR Presentation and Document:	20%
TRR Presentation and Test Documentation:	20%
Final Report:	20%
Final Video:	10%
Progress Grades:	10%

Grading

Final grades will be assigned based upon completion of the major deliverables and the progress scores according to the following weights.

In general, we will follow the 90% / 80% / 70% / 60% scale with +/- grades. Depending on class performance, the instructor reserves the right to curve the scale, if necessary, based on his discretion. The scale will never be curved downward, against the class.

Peer-review feedback will be submitted at three points throughout the semester and will be taken into consideration when assigning final, individual grades.

Other Notes and Policies

Questions

The instructor is available to extend your access to help in this course outside of class hours. Please maximize the group work and laboratory times to interface with the instructor.

Attendance

The University Regulations Handbook stated that "Students are expected to be present for every meeting of the classes in which they are enrolled." Regular attendance will not be taken but if you must miss a class: 1) you must inform the instructor (by email) and 2) you are responsible for the lecture material, assignments and/or announcements made.

Changes and Emergencies

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructor via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

Copyright

Among the materials that may be protected by copyright law are the lectures, notes, and other material presented in class or as part of the course. Always assume the materials presented by an instructor are protected by copyright unless the instructor has stated otherwise. Students enrolled in, and authorized visitors to, Purdue University courses are permitted to take notes, which they may use for individual/group study or for other non-commercial purposes reasonably arising from enrollment in the course or the University generally.

Class videos, notes, and handouts are considered to be 'derivative works' of the instructor's presentations and materials, and they are thus subject to the instructor's copyright. As such, they should not be uploaded to any share site, sold or bartered without express written permission. To be clear: this means that uploading any course material on 'CourseHero', ChatGPT, Copilot (even the Purdue branded version) or similar websites is a violation of the syllabus and a violation of copyrights.

Use of artificial intelligence (AI) or Large Language Models (LLM) in this course

AIAA has established a policy for the use of AI when publishing with AIAA. (See here) The following policy is closely aligned with AIAA's policy and provides you with an industry relevant AI policy such that you are better prepared for your future career. This policy addresses the rapidly growing trend of using artificial intelligence (AI) technology in the preparation of written materials (this includes all presentations and reports in this class). When this document refers to AI, it includes all forms of AI. This term is inclusive of, but not limited to, intelligent agents, generative or creative AI (such as ChatGPT and openFrameworks), AI-assisted technologies, deep learning, etc. This policy is focused on the use of AI in the preparation of text, figures, illustrations, references, and graphics for a presentation or a report. AI tools used in the design activity are not addressed in this policy, but they are expected to be fully described in the presentation or report for evaluation by the teaching team. This policy is intended to provide guidance and transparency to students, instructors, teaching assistants, and graders concerning how AI should be ethically used in this class.

Each student is fully responsible and accountable for the content of the entire presentation and report, including the accuracy, integrity, and completeness, citation of previous work, obtaining copyright approval, and potential issues of plagiarism. If the students use any type of AI technology in the writing process, AI should be used primarily to improve readability, grammar, and language in the work. Students may not list AI or AI-assisted technologies as a co-author. Students may not cite AI engines as an original source. When students use AI in the writing or presentation process, they must disclose the use of AI technologies used in the preparation of the presentation or report upon presentation/submission to the teaching team; disclosures exclude the use of spelling and grammar checkers that are included in word processing software.

AI can be used to construct or modify illustrations, figures or graphics to improve clarity and the communication of concepts. AI can be used to modify a photograph to improve the clarity or particular features of interest. However, AI shall not be used to generate false or misleading photographs, such as adding objects or images of individuals. Regardless of how an illustration, figure, graphic, or photograph is constructed, the students are fully responsible for appropriate citation to previous work, obtaining needed copyright permission, and potential issues of plagiarism. When students use AI for an illustration, figure, graphic, or photograph, they must disclose the use of AI technologies used when the presentation or report is presented/submitted to the teaching team. If AI is used in the writing process or figure construction as permitted above, the students must include a brief description of AI use in the presentation or report.

A written document submitted to the teaching team (the instructor, teaching assistant(s), and grader(s)) is treated as a confidential document. The teaching team may not use AI to generate a review of the written document. The teaching team can use AI as a supplement to their personal expertise, e.g., to find related publications in the written document subject areas, though the teaching team is responsible for confirming all information used during the review that is not obtained from peer-reviewed sources. The teaching team will not upload a submitted written document, or any portion of it, to an AI tool or website. The teaching team can use software provided by Purdue University, such as iThenticate, that compares a submitted written document with previously published or submitted works to aid in detecting similar works and potential plagiarism. Any software used to detect similar works or potential plagiarism, other than provided by Purdue University, will be identified by the teaching team.